

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: B. Tech.

Branch : Common To All Branches

Semester :II

Subject Code & Name: BTBS202 Engineering Chemistry

Max Marks: 60

Date:02/07/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Water containing $Mg(HCO_3)_2$ & $CaCl_2$ is -----	2	1
	a. Permanent hardness only b. Temporary hardness only c. Both a & b d. None of these		
2	Temporary hardness can be removed by -----	2	1
	a. EDTA method b. Mohr's method c. Boiling d. Precipitation		
3	The colour of metal-EBT complex is -----	2	1
	a. Blue b. Wine red c. Pink d. Colourless		
4	A system consisting of ice and water is -----	2	1
	a. One phase system b. Two phase system c. Three phase system d. None of these		
5	For the Sulphur system no. of component is -----	2	1
	a. 3 b. 1 c. 2 d. None of these		
6	During electrochemical corrosion in alkaline environment:	2	1
	a. oxygen evolution occurs b. oxygen absorption occurs c. hydrogen evolution takes place d. hydrogen absorption takes place		

7	If the metal is more pure then rate of corrosion is very ---				2	1
	a. Less	b. More	c. High	d. None of these		
8	Galvanizing is the process of coating -----over the protective metal.				2	1
	a. Sn	b. Zn	c. Cu	d. Ni		
9	Fuel is a combustible substance, containing ----- as a main constituent.				1	1
	a. Hydrogen	b. Oxygen	c. Nitrogen	d. Carbon		
10	Oiliness is least in case of -----				2	1
	a. Greases	b. Mineral oils	c. Animal oils	d. Palm oils		
11	The conductivity of an electrolyte is due to the-----				1	1
	a. Presence of impurity in the electrolyte	b. Free movement of ions in the solution	c. Reunion of ions in the solution	d. Release of heat energy due to ionization		
12	Methyl orange has pH range is _____				1	1
	a. 4.3 - 5.4	b. 3.2 - 4.5	c. 4.4 - 5.0	d. 4.1 - 5.3		
Q. 2	Solve the following.					12
A)	Explain Ion exchange process of softening of water with its advantages and disadvantages.				2	6
B)	Describe the determination of dissolved oxygen in water by Iodometric method.				2	6
Q.3	Solve the following.					12
A)	Explain the Phase diagram of two component Ag-Pb system.				2	6
B)	Explain Gibbs Phase Rule with 1) Degree of freedom 2) No. of Component terminologies.				2	6
Q. 4	Solve Any Two of the following.					12
A)	Explain oxygen absorption mechanism of wet corrosion.				2	6
B)	Explain factors influencing rate of corrosion.				1	6

C)	Explain how corrosion of metals controlled by Anodic protection method.	2	6
Q.5	Solve Any Two of the following.		12
A)	Explain any six characteristics of good fuel in detail.	1	6
B)	Explain how the percentage of moisture and volatile matter in coal can be estimated.	2	6
C)	Describe in brief solid and semisolid lubricants.	2	6
Q. 6	Solve Any Two of the following.		12
A)	Write note on: Glass electrode.	1	6
B)	Explain Ostwald's theory of acid – base indicator.	2	6
C)	Explain conductometric titration with suitable example.	2	6
	*** End ***		